16 There are only red counters and blue counters in a bag.

Joe takes at random a counter from the bag.
The probability that the counter is red is 0.65
Joe puts the counter back into the bag.

Mary takes at random a counter from the bag.
She puts the counter back into the bag.

(a) What is the probability that Joe and Mary take counters of different colours?

P(blue) = 1 – 0.65 = 0.35

P(red then blue) = 0.65 × 0.35 = 0.2275

P(blue then red) = 0.35 × 0.65 = 0.2275

1 mark
Multiply for one order

P(different) = 0.2275 + 0.2275 = 0.455

1 mark
Add both orders

0.455

(2)

There are 78 red counters in the bag.

(b) How many blue counters are there in the bag?

	Percent	Counters	
÷13	65	78	÷13
×7	5	6	×7
	35	42	

1 mark
Use 65% = 78 counters

1 mark
35% = 42 blue counters

42

(2)

(Total for Question 16 is 4 marks)